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Hypertension

ORIGINAL ARTICLE

Timing of Antihypertensive Drug Therapy: A Systematic Review and Meta-Analysis of Randomized Clinical Trials

Muhammad Haisum Maqsood,* Franz H. Messerli,* Adam H. Skolnick, Jonathan D. Newman, Jeffrey S. Berger, Sripal Bangalore

BACKGROUND: The timing of antihypertensive drugs administration is controversial. The aim was to compare the efficacy of dosing of antihypertensive drugs in the morning versus evening.

METHODS: A PubMed, EMBASE, and clinicaltrials.gov databases search for randomized clinical trials of antihypertensive therapies where patients were randomized to morning versus evening dosing. The outcomes were ambulatory blood pressure (BP) parameters (day-time, night-time, and 24/48-hour systolic blood pressure [SBP] and diastolic blood pressure [DBP]) and cardiovascular outcomes.

RESULTS: Of 72 randomized controlled trials included, evening dosing significantly reduced ambulatory BP parameters: 24/48-hour SBP (mean difference [MD]=1.41 mm Hg; [95% CI, 0.48–2.34]), DBP (MD=0.60 mm Hg [95% CI, 0.12–1.08]), night-time SBP (MD=4.09 mm Hg [95% CI, 3.01–5.16]), DBP (MD, 2.57 mm Hg [95% CI, 1.92–3.22]), with a smaller reduction in day-time SBP (MD=0.94 mm Hg [95% CI, 0.01–1.87]), and DBP (MD=0.87 mm Hg [95% CI, 0.10–1.63]), and numerically lower cardiovascular events compared with morning dosing. However, when controversial data by Hermida (23 trials, 25 734 patients) were omitted ($P_{heterogeneity}$<0.05 for most outcomes), the above effect of evening dosing attenuated with no significant effect on 24/48-hour ambulatory blood pressure, day-time BP, and major adverse cardiac event and smaller reduction in night-time ambulatory SBP and DBP.

CONCLUSIONS: Evening dosing of antihypertensive drugs significantly reduced ambulatory BP parameters and lowered cardiovascular events but the effect was mainly driven by trials by Hermida group. Unless the intention is to specifically lower night-time BP, antihypertensive drugs should be taken at a time of day that is convenient, optimizes adherence, and minimizes undesirable effects. ***(Hypertension.* 2023;80:1544–1554. DOI: 10.1161/HYPERTENSIONAHA.122.20862.)** • **Supplement Material**.

Key Words: adverse cardiovascular events ■ ambulatory blood pressure ■ chronotherapy ■ evening dosing ■ hypertension ■ morning dosing

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Antihypertensive drugs are some of the most commonly prescribed medications in the United States and worldwide.[1] However, the timing of their administration remains controversial,[2–5] and society guidelines have been silent on this issue.[6,7] Traditionally, antihypertensive drugs are administered in the morning. One reason for morning dosing may be that adherence with the antihypertensive regime has been documented to diminish with evening dosing.[8,9] This morning tradition has been challenged; for almost 2 decades multiple data from a single center suggested improved blood pressure (BP) control and cardiovascular outcomes with evening dosing.[4,10–33] In contrast, a recently reported prospective randomized trial of 21 104 patients, randomized to morning

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NOVELTY AND RELEVANCE

What Is New?

This systematic review and meta-analysis of 72 randomized controlled trials evaluated the effect of antihypertensive drugs morning versus evening dosing and showed significant reduction of ambulatory blood pressure (BP) and numerically lower risk of major adverse cardiac event, cardiovascular mortality, all-cause mortality, and heart failure in patients randomized to evening dosing when compared with morning. When controversial single-center data by Hermida et al were omitted, evening dosing of antihypertensive drugs significantly reduced night-time ambulatory BP but had no effect on other ambulatory BP parameters and on cardiovascular outcomes.

What Is Relevant?

BP reduction with antihypertensive agents reduces cardiovascular events. However, the optimal timing of medication administration is unknown. The trials conducted on the efficacy of antihypertensive drugs morning versus evening dosing were small-scale and had heterogeneous outcomes. Particularly, multiple data from a single center have suggested improved 24/48-hour BP control and outcome with evening dosing. Evening dosing of antihypertensive drugs significantly reduce night-time BP when compared with morning dosing but has no certain effect on other ambulatory BP parameters and on cardiovascular outcomes, but results were largely driven by trials conducted by Hermida and colleagues.

Clinical/Pathophysiological Implications?

Unless the intention is to specifically lower nighttime BP, antihypertensive drugs should be taken at a time of day that is convenient, optimizes adherence and minimizes undesirable effects.

Nonstandard Abbreviations and Acronyms

BP	blood pressure
DBP	diastolic blood pressure
MACE	major adverse cardiac event
MD	mean difference
RCT	randomized controlled trial
SBP	systolic blood pressure

dosing versus evening dosing, found that heart attack, stroke, and vascular death occurred to a similar proportion regardless of the time of administration.[9,34] This led to a concensus document which concluded that evening dosing should not be preferred.[35] Numerous other trials on the efficacy of morning versus evening dosing of antihypertensive drugs have been conducted. Further, population-derived data suggest that night-time BP is a better predictor of mortality in hypertensive patients.[36,37] The aim of this study was to compare the efficacy of antihypertensive drugs evening versus morning dose by a meta-analysis of randomized trials.

METHODS

Data Availability

This meta-analysis and systematic review was conducted in accordance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.[38] This analysis was not considered for institutional review board review as this was a trial level meta-analysis of published clinical trial data. The protocol of this meta-analysis was registered in PROSPERO (CRD42022333583). The data that support the findings of this study are available from the corresponding author upon reasonable request.

Data (collection sheets) would be made available at a reasonable request.

Search Strategy and Selection Criteria

A time limited search until August 26, 2022 was conducted using PubMed and EMBASE databases. The following Medical Education Subject Headings were used for this search: "chronotherapy," "conventional therapy," "antihypertensives," "hypertension," "morning dose," "evening dose," "adverse cardiovascular events," "night time blood pressure," "day time blood pressure," and "post-prandial dose." To identify grey literature, www.clinicaltrials.gov was searched. We also considered references of the eligible studies to identify further studies.

The inclusion criteria were randomized controlled trials (RCTs) that compared the effect of morning versus evening dosing of antihypertensive drugs on changes in ambulatory BP parameters (24/48-hour, night-time, and day-time ambulatory systolic blood pressure [SBP] and diastolic blood pressure [DBP]) and clinical outcomes. Studies with a placebo comparison arm (eg, morning versus placebo and evening versus placebo) or without a comparison arm, observational studies, editorials, letters, and review articles were excluded.

We used PICO (Population, Intervention, Comparison, and Outcome) criteria for study selection.[39] Population of interest: hypertensive patients who were enrolled in antihypertensive trials of morning versus evening dosing; intervention: antihypertensive drug evening dosing; comparison: antihypertensive drug morning dosing; outcomes: reduction in ambulatory BP.

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Study Outcomes

The efficacy outcomes were reduction in ambulatory SBP and DBP at day-time, night-time, and 24/48-hour BP in mm Hg. Ambulatory 24/48-hour BP was based on either 24- or 48-hour BP measurements. Clinical outcomes were major adverse cardiac event (MACE; defined as a composite of angina, myocardial infarction, heart failure, vascular interventions, stroke, cardiovascular and all-cause mortality), all-cause mortality, cardiovascular mortality, and heart failure.

Data Analysis

Data Extraction and Quality Assessment

The studies were screened by 2 independent authors (M.H.M. and S.B.). Nonrelevant studies were excluded based on title followed by abstract. Full text studies were then screened for final selection on basis of above-mentioned, prespecified inclusion criteria. Two reviewers (M.H.M. and S.B.) independently reviewed the literature, extracted and entered data. Discrepancies were resolved by discussion. Two authors independently appraised the methodological quality of included trials using the Cochrane Risk of Bias Tool which assessed selection, allocation, performance, detection, attrition, and reporting bias (Table S1).[40]

Statistical Analysis

Continuous variables were reported as mean with SD and categorical variables were expressed as frequency/percentage. Mean difference (MD) with 95% CIs and adverse events using odds ratio (OR), 95% CIs were calculated using random effects model and DerSimonian-Laird method.[41] Heterogeneity between studies was assessed using I^2 statistic.[42] I^2 value <25%, 25% to 50%, 50% to 75%, and >75% indicated low, moderate, high and extreme heterogeneity, respectively. Subgroup analysis was conducted based on Hermida versus non-Hermida et al. because for almost 2 decades multiple data from their single center suggested improved BP control and cardiovascular outcome with evening dosing. A sensitivity analysis was done by including trials with at least 50 subjects enrolled. The effect of study size was evaluated using cumulative meta-analysis starting from largest to smallest study. Meta-regression analysis was performed to evaluate the relationship of year of publication, percentage men, and mean age on outcomes. Funnel plots and eggers' regression tests were used to assess publication bias. Leave-one-out analysis was conducted to see if individual study might have had a disproportionate influence on the final results. A $P<0.05$ was considered statistically significant. Risk estimates and effect sizes, sensitivity analyses, and cumulative meta-analyses were calculated using Stata version 17.0 software (Stata Corporation, College Station, TX).

RESULTS

The initial search yielded 1718 reports, which were screened by reviewing the title or abstract of which 72 RCTs were included in the final meta-analysis (Figure S1).

Baseline Characteristics

Of the 72 RCTs, 69 evaluated ambulatory BP effect enrolling 14964 patients randomized to evening dosing and 14969 to morning dosing group. Of the included trial 32% [22/69]) were cross-over trials. The mean age was 59 years with 54% of randomized participants being men. The doses of antihypertensive drugs between morning and evening dosing were similar. Among the included trials, 33% (23 out of 69) trials were by Hermida and colleagues and consisted of 25734 patients (12853 evening versus 12899 morning; 1 cross-over trial). Similarly, there were only 8 out of 69 (11.6%) trials where the antihypertensive trials used was short-acting of which 3 were by Hermida et al.

Six trials were considered for evaluation of clinical outcomes with a total of 59976 patients of which 29868 were randomized to evening dosing and 30108 to morning dosing group. Of these 6 RCTs, 4 had similar doses of antihypertensive drugs between morning and evening timings. The 6 included trials enrolled patients with similar proportion of men (52.3%) and women, mean age 63.3 years and were followed up for a mean of 58.6 months (4.8 years). Among the included trials, 50% (3 out of 6) trials were by Hermida and colleagues and consisted of 22016 patients (10995 evening versus 11021 morning). The baseline characteristics of the study population are provided in Table S2, and characteristics of Hermida and non-Hermida studies are shown in Table 1. Ambulatory BP measurement parameters criteria for day-time, night-time, 24 or 48 hours mean, and the methodology for the 24/48 hours BP measurement in the included trials are shown in Table S3.

Outcomes

24/48-Hour Ambulatory BP

Evening dosing led to greater reduction in 24/48-hour ambulatory SBP (MD=1.41 mm Hg [95% CI, 0.48–2.34], I^2=82%; 53 trials) compared with morning dosing (Figure 1A). Sensitivity analysis restricted to trials consisting of at least 50 subjects showed similar result (SBP: MD=1.79 mm Hg [95% CI, 0.69–2.89], I^2=88%; 34 trials; Figure S2A). Subgroup analysis based on Hermida versus non-Hermida trials ($P_{heterogeneity}$=0.01) showed significant BP lowering effect with evening dosing only in the trials by Hermida et al (MD=2.30 mm Hg [95% CI, 0.90–3.70]; I^2=92%) but not in the non-Hermida trials (MD=0.16 mm Hg [95% CI, −0.56 to 0.87], I^2=0%; Figure 1A).

Evening dosing led to greater reduction in 24/48-hour ambulatory DBP (MD=0.60 mm Hg [95% CI, 0.12–1.08], I^2=57%%; 54 trials) compared with morning dosing (Figure 1B). Sensitivity analysis restricted to trials consisting of at least 50 subjects showed similar result (DBP: MD=0.75 mm Hg [95% CI,0.19–1.31], I^2=70%; 34 trials]; Figure S3A). Subgroup analysis of Hermida versus non-Hermida trials ($P_{heterogeneity}$=0.01) showed significant BP lowering effect with evening dosing only in the trials by Hermida et al (MD=0.97 mm Hg [95% CI, 0.30 to 1.64], I^2=77%) but not in the non-Hermida trials

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Table 1. Summary Characteristics of Hermida and Non-Hermida Trials

	Total	Hermida data	Non-Hermida data
Blood pressure parameters			
Number of studies	69	23	46
Number of patients	29 265	25 734	3531
Number of RCTs	69	23	46
*Number of studies favoring PM dosing for 24/48-hour SBP	11/53	10/21	1/32
Clinical outcomes			
Number of studies	6	3	3
Number of patients	59 976	22 016	37 960
Number of RCTs	6	3	3
*Number of studies favoring PM dosing for MACE	3	3	0
Antihypertensive medications tested†			
ACE/ARBs as single drug	23	10	13
Beta-blockers as single drug	4	1	3
CCBs as a single drug	13	5	8
Combined CCBs and ACE/ARBs only	8	1	7

ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; CCB, calcium channel blocker; MACE, major adverse cardiac event; PM, evening dose; RCT, randomized controlled trial; and SBP, systolic blood pressure.

*Taken from forest plot if null value is not within the 95% CI of mean value.

†Antihypertensive medications, which were the only intervention as morning or evening dose.

(MD=−0.21 mm Hg [95% CI, −0.75 to 0.33], I^2=0%; Figure 1B).

Night-Time Ambulatory BP

Evening dosing led to greater reduction in night-time SBP (MD=4.09 mm Hg [95% CI, 3.01–5.16], I^2=86%; 65 trials) compared with morning dosing (Figure 2A). Sensitivity analysis restricted to trials consisting of at least 50 subjects showed similar result (SBP: MD=4.43 mm Hg [95% CI, 3.25–5.61], I^2=90%; 41 trials; Figure S4A). Subgroup analysis showed no significant heterogeneity of treatment effect based on Hermida versus non-Hermida trials ($P_{heterogeneity}$=0.35) but the reduction in night-time SBP with evening dosing was smaller in the non-Hermida trials (Figure 2A).

Evening dosing led to greater reduction in night-time DBP (MD=2.57 mm Hg [95% CI, 1.92–3.22], I^2=72%; 57 trials) compared with morning dosing (Figure 2B). Sensitivity analysis restricted to trials consisting of at least 50 subjects showed similar result (DBP: MD=2.92 mm Hg [95% CI, 2.22–3.63], I^2=79%; 36 trials; Figure S5A). Subgroup analysis based on Hermida versus non-Hermida trials ($P_{heterogeneity}$≤0.001) greater magnitude of BP lowering effect with evening dosing in the trials by Hermida et al (MD=3.68 mm Hg [95% CI, 2.81–4.54], I^2=86%) when compared with the non-Hermida trials (MD=0.97 mm Hg [95% CI, 0.18–1.76], I^2=13%; Figure 2B).

Day-Time Ambulatory BP

Evening dosing of antihypertensive drugs led to greater reduction in day-time SBP compared with morning dosing but the magnitude was small (MD=0.94 mm Hg [95% CI, 0.01–1.87]; I^2=81%; 66 trials; Figure 3A). Sensitivity analysis restricted to trials consisting of at least 50 subjects showed similar result (MD=1.30 mm Hg [95% CI, 0.19–2.42), I^2=88%; 41 trials; Figure S6A). Subgroup analysis based on Hermida versus non-Hermida trials ($P_{heterogeneity}$=0.78) showed similar effect of evening dosing without significant heterogeneity of treatment effect (Figure 3A).

Evening dosing led to a greater reduction in day-time DBP compared with morning dosing but the magnitude was small (MD=0.87 mm Hg [95% CI, 0.10–1.63]; I^2=85%; 60 trials; Figure 3B). Sensitivity analysis restricted to trials consisting of at least 50 subjects showed similar result (DBP: MD=1.03 mm Hg [95% CI, 0.08–1.98]; I^2=91%; 35 trials; Figure S7A). Subgroup analysis showed no significant heterogeneity of treatment effect based on Hermida versus non-Hermida trials ($P_{heterogeneity}$=0.91; Figure 3B).

There was no small study effect (Figures S2B through S7B) or evidence of publication bias (Figures S2C through S7C) in any of the above analyses. Meta-regression analysis did not show any difference in outcome based on year of publication, percentage men, and mean age for any of the above analyses except for night-time DBP (*P*=0.019; Figure S2D through S2F, Figure S3D through S3F, Figure S4D through S24F, Figure S5D through S5F, Figure S6D through S6F, Figure S7D through S7F). Leave-one-out analysis did not show any influence on final outcomes for 24/48 hours and night-time ambulatory BP but showed influence of several studies on day-time ambulatory BP (Figures S2G through S7G).

Clinical Outcomes

Risk of MACE (OR=0.68 [95% CI, 0.46–1.01]; I^2=96%; *P*=0.06; 6 trials; Figure 4A), cardiovascular mortality (OR=0.47 [95% CI, 0.21–1.04]; I^2=92%; *P*=0.06; 4 trials; Figure 4B), all-cause mortality (OR=0.64 [95% CI, 0.37–1.08], I^2=93%; *P*=0.10; 5 trials; Figure 4C), and heart failure (OR=0.54 [95% CI, 0.28–1.02], I^2=91%; *P*=0.06; 4 trials; Figure 4D) were numerically lower with evening compared with morning dosing, and reached statistical significance in a sensitivity analysis, which excluded trials with different evening and morning antihypertensive drug doses MACE (OR=0.50 [95% CI, 0.32–0.79], I^2=95%; *P*≤0.001; 4 trials; Figure S8A), cardiovascular mortality (OR=0.39 [95% CI, 0.30–0.49], I^2=0%; *P*≤0.001; 3 trials; Figure S8B), all-cause mortality (OR=0.50 [95% CI, 0.43–0.57], I^2=0%; *P*≤0.001; 3 trials; Figure S8C), and heart failure (OR=0.39 [95% CI, 0.21–0.72], I^2=69%; *P*≤0.001; 3 trials; Figure S8D). Subgroup analysis based on Hermida versus non-Hermida trials

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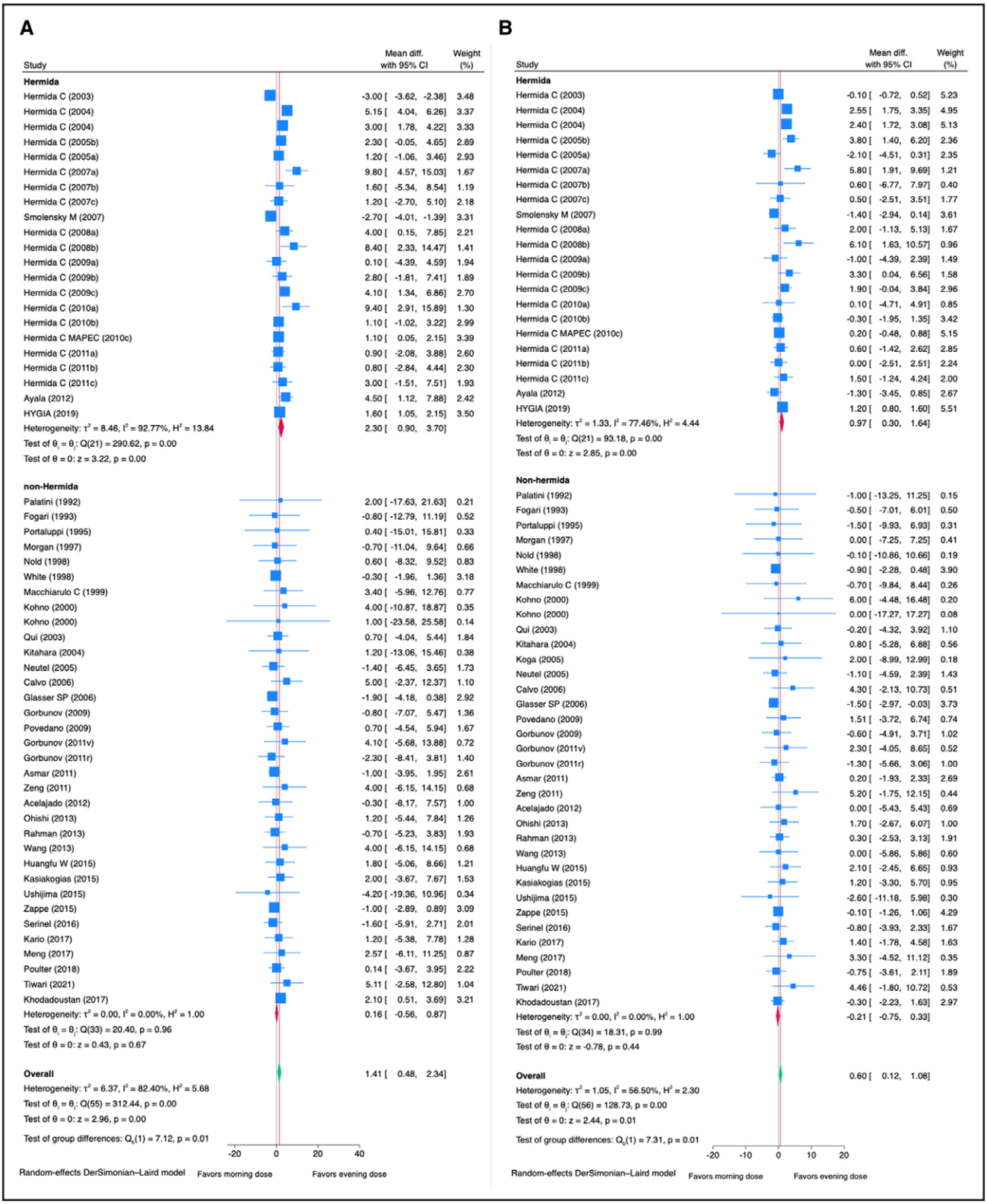


Figure 1. The 24/48-hour ambulatory blood pressure.
A, Systolic blood pressure (SBP) and (**B**) diastolic blood pressure (DBP). Hermida et al (2004)[12] is a single trial with 2 different antihypertensive groups (mono-therapy and polytherapy), which were analyzed separately. Kohno et al (2000)[43] is a single trial with 2 different groups (dippers vs nondippers) analyzed separately. Gorbunov et al (2011)[44] is a single trial with 2 different antihypertensives used (ramipril and verapamil) analyzed separately.

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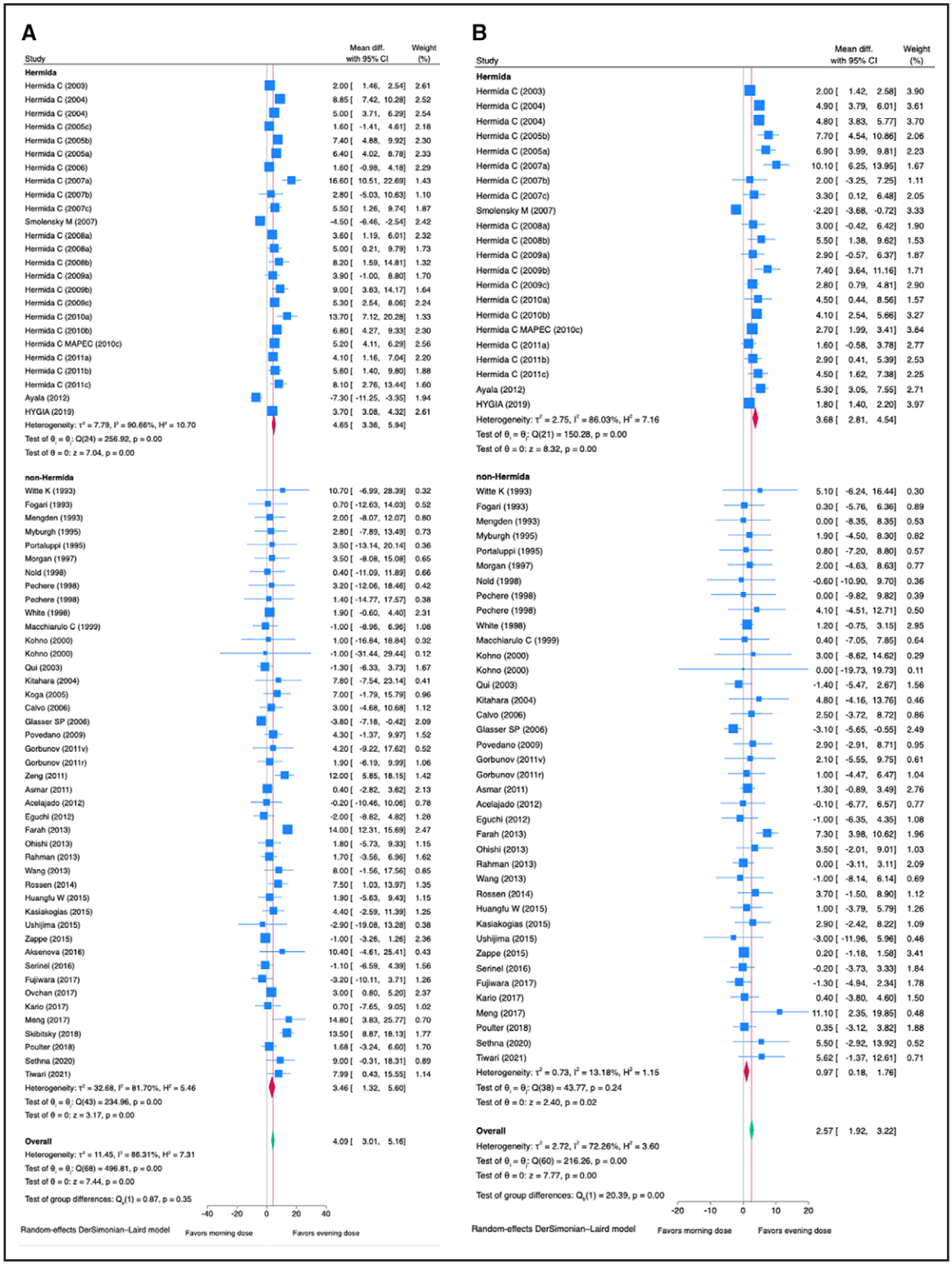


Figure 2. Night-time ambulatory blood pressure.
A, Systolic blood pressure (SBP) and (**B**) diastolic blood pressure (DBP). Hermida et al (2004)[12] is a single trial with 2 different antihypertensive groups (mono-therapy and polytherapy), which were analyzed separately. Kohno et al (2000)[43] is a single trial with 2 different groups (dippers vs nondippers) analyzed separately. Pechère-Bertschi et al (1998)[45] is a single trial with 2 different antihypertensives used (irbesartan and enalapril) analyzed separately. Gorbunov et al (2011)[44] is a single trial with 2 different antihypertensives used (ramipril and verapamil) analyzed separately.

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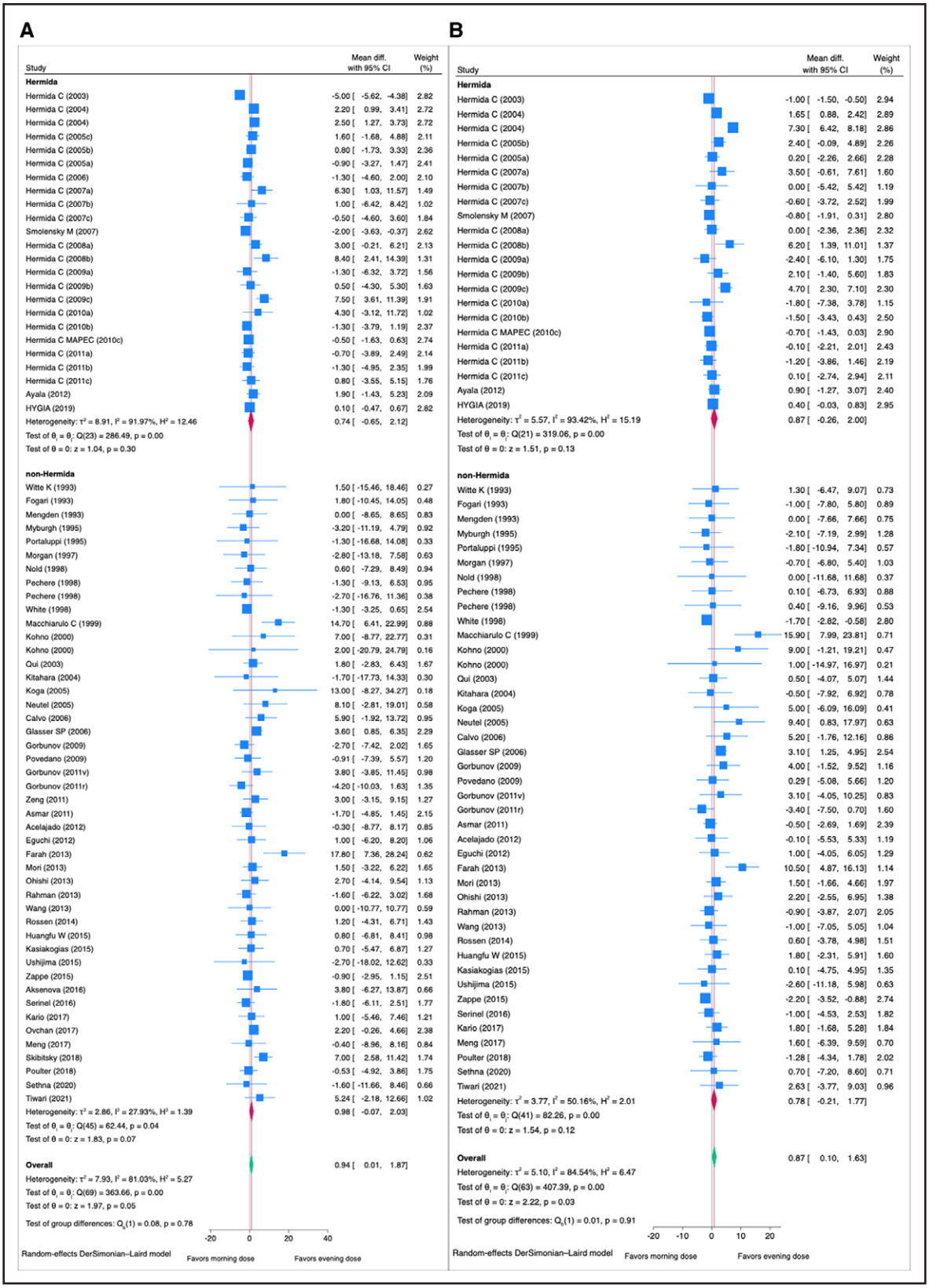


Figure 3. Day-time ambulatory blood pressure.

A, Systolic blood pressure (SBP) and (**B**) diastolic blood pressure (DBP). Hermida et al (2004)[12] is a single trial with 2 different antihypertensive groups (mono-therapy and polytherapy), which were analyzed separately. Kohno et al (2000)[43] is a single trial with 2 different groups (dippers vs nondippers) analyzed separately. Pechère-Bertschi et al (1998)[45] is a single trial with 2 different antihypertensives used (irbesartan and enalapril) analyzed separately. Gorbunov et al (2011)[44] is a single trial with 2 different antihypertensives used (ramipril and verapamil) analyzed separately.

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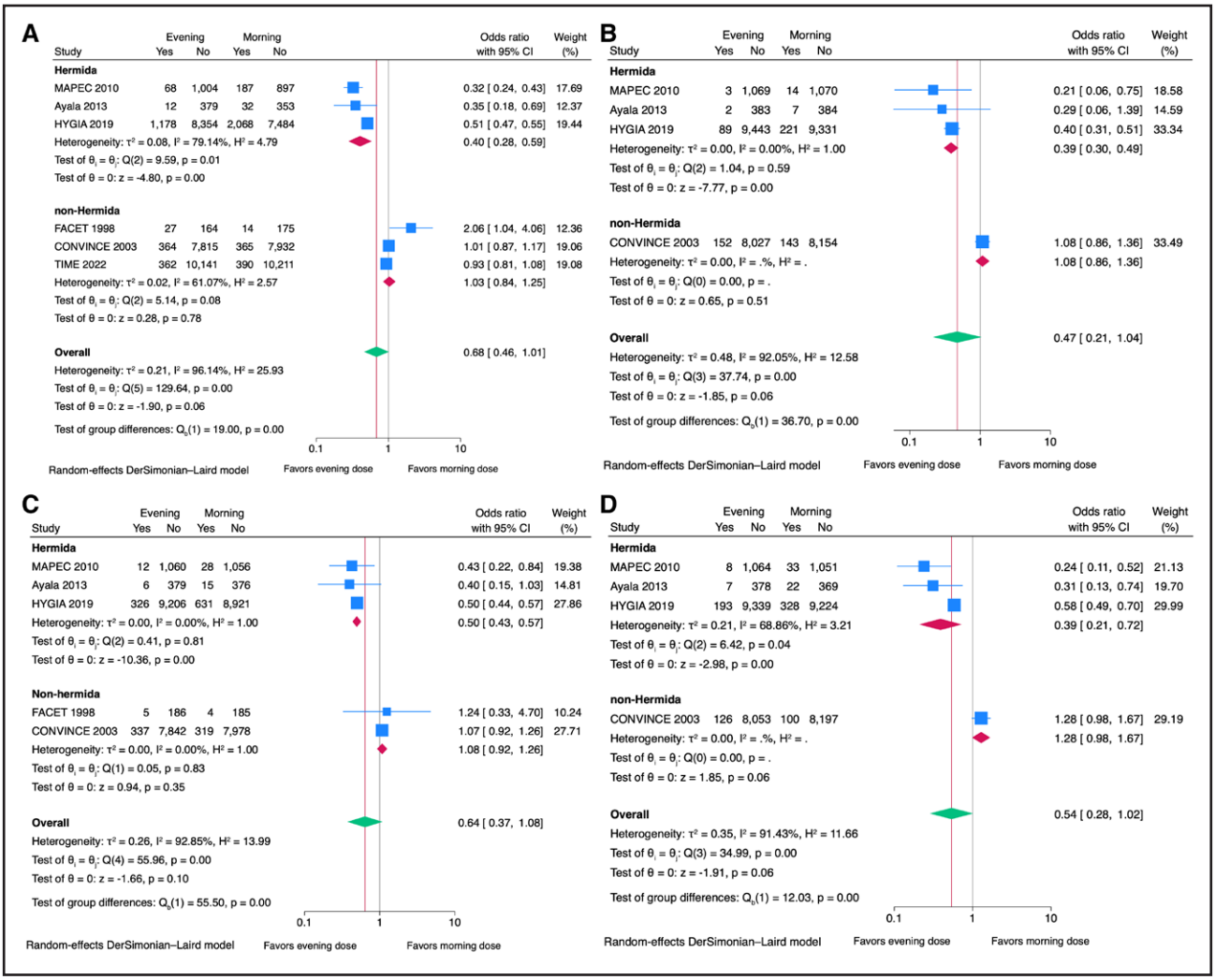


Figure 4. Clinical outcomes with evening compared to morning dose.
A, Major adverse cardiac event (MACE), (**B**) cardiovascular mortality, (**C**) all-cause mortality, and (**D**) heart failure.

($P_{heterogeneity}$ <0.001) showed significantly lower MACE, cardiovascular mortality, all-cause mortality, and heart failure with Hermida trials only with no significant difference in outcomes with non-Hermida trials (Figure 4A through 4D). Leave-one-out analysis showed that omission of CONVINCE trial showed significant difference in all 4 outcomes (P<0.05; Figure S9A through S9D).[46] There was no evidence of publication bias with Egger's test (MACE P=0.78, all-cause mortality P=0.88, cardiovascular mortality P=0.22, and heart failure P=0.085). Summary of outcomes is shown in Table 2.

DISCUSSION

This meta-analysis of 72 RCTs evaluating the effect of antihypertensive drugs morning versus evening dosing showed significant reduction of ambulatory BP, and numerically lower risk of cardiovascular events in patients randomized to evening dosing when compared with morning dosing. However, this beneficial effect on BP and cardiovascular outcomes were exclusively contingent on controversial single-center data.[4,10–33] Hence, the present data confirm the outlier status of the data of Hermida et al. Subgroup analysis excluding trials by Hermida et al showed smaller but significant reduction in night-time ambulatory BP with evening dosing when compared with morning dosing but no significant difference in other BP parameters or on clinical outcomes.

Traditionally antihypertensive drugs are administered in the morning.[4] However, multiple trials from the single-center studies by Hermida et al have suggested improved BP control and outcome with evening dosing.[4,10–33] For almost 2 decades, numerous concerns were voiced about these data, particularly also about the 2 randomized controlled trials by Hermida et al, MAPEC and HYGIA.[2,3,47,48] These considerations led to a letter of concern issued by European Heart Journal in 2020,[49] which could not document scientific misconduct but outlined concerns regarding some of the findings.[50]

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Table 2. Summary of Outcomes

Outcome	Total	Hermida trials	Non-Hermida trials	*P* value of difference in groups ($P_{heterogeneity}$)
Blood pressure metrics				
	Mean difference with 95% CI			
Day-time ambulatory SBP change, mm Hg	0.94 (0.01 to 1.87)	0.74 (−0.65 to 2.12)	0.98 (−0.07 to 2.03)	0.78
Day-time ambulatory DBP change, mm Hg	0.87 (0.10 to 1.63)	0.87 (−0.26 to 2.0)	0.78 (−0.21 to 1.77)	0.91
Night-time ambulatory SBP change, mm Hg	4.09 (3.01 to 5.16)	4.65 (3.36 to 5.94)	3.46 (1.32 to 5.60)	0.35
Night-time ambulatory DBP change, mm Hg	2.57 (1.92 to 3.22)	3.68 (2.81 to 4.54)	0.97 (0.18 to 1.76)	<0.001
The 24/48-hour ambulatory SBP change, mm Hg	1.41 (0.48 to 2.34)	2.30 (0.90 to 3.70)	0.16 (−0.56 to 0.87)	0.01
The 24/48-hour ambulatory DBP change, mm Hg	0.60 (0.12 to 1.08)	0.97 (0.30 to 1.64)	−0.21 (−0.75 to 0.33)	0.01
Clinical outcome metrics				
	Odds ratio with 95% CI			
MACE	0.68 (0.46 to 1.01)	0.40 (0.28 to 0.59)	1.03 (0.84 to 1.25)	<0.001
Cardiovascular mortality	0.47 (0.21 to 1.04)	0.39 (0.30 to 0.49)	1.08 (0.86 to 1.36)	<0.001
All-cause mortality	0.64 (0.37 to 1.08)	0.50 (0.43 to 0.57)	1.08 (0.92 to 1.26)	<0.001
Heart Failure	0.54 (0.28 to 1.02)	0.39 (0.21 to 0.72)	1.28 (0.98 to 1.67)	<0.001

DBP indicates diastolic blood pressure; MACE, major adverse cardiac event; and SBP, systolic blood pressure.

Our data document that evening dosing compared to morning dosing lowers night-time SBP by about 4 mm Hg (4.6 mm Hg in Hermida trials and 3.4 mm Hg in the non-Hermida trials). This nocturnal BP fall is clinically meaningful and, depending on the situation, as Burnier et al[47] have outlined, may possibly confer benefits or harm.[47] Clearly, it should be carefully taken into account when evening dosing is considered and these results are in line with a recent consensus document by Stergiou et al.[35]

Most recently, the TIME (Treatment in Morning versus Evening) study from the United Kingdom which randomized 21 104 patients (10 503 in evening dose group and 10 601 in morning dose group) with a median follow-up of 5.2 years reported no significant differences (unadjusted hazard ratio 0.95 [95% CI, 0.83–1.10]; *P*=0.53) in primary composite end point (hospitalization for nonfatal myocardial infarction or nonfatal stroke, or vascular death) between the 2 groups.[9,34,51] The TIME study clearly confirms the outlier status of the single center Hermida data. Apart from wanting to achieve a slightly lower nocturnal BP level, there currently is no valid clinical reason to switch to bedtime dosing of antihypertensive drugs. To the contrary, evening dosing when compared with morning dosing has been shown to reduce drug adherence.[52] By electronic monitoring, Vrijens et al[8] showed that morning takers were significantly more likely to take treatment correctly than evening takers (OR=1.38 with 95% CI 1.36 to 1.41). Similarly, the TIME study documented nonadherence at anytime of 22.5% with morning dosing as opposed to 39% with evening dosing (*P*<0.001).[9] Results from the ongoing randomized cardiovascular outcome trial Bed-Med from Canada will further add to present data.[53] Hopefully, the evidence in aggregate will now suffice to motivate future guidelines to make firm recommendations. In the broad context, with the use of long-acting antihypertensive drugs, the issue of night-time dosing is likely less relevant.

Study Limitations

The results of this analysis should be interpreted considering certain limitations. First, our analysis consisted mainly of small-scale RCTs in specific geographic areas so the extent of generalizability and trends in different populations is unknown. Second, adherence with antihypertensive drugs was not reported in most of these trials. Third, not all commercially available antihypertensive drugs were represented in this meta-analysis. Fourthly, there was variability in drug administration between trials with few trials designed to give all of the antihypertensive medications in the evening while in others only a few antihypertensive medications were given in the evening. Finally, there is a potential for overlap of patients in different Hermida trials, which could artificially increase the benefits of evening dose.

CONCLUSIONS

Evening dosing of antihypertensive drugs significantly reduced ambulatory BP and lowered cardiovascular outcomes but the effects were largely driven by trials by Hermida et al. When these trials were omitted, evening dosing reduced night-time ambulatory BP when compared with morning dosing but had no certain effect on other ambulatory BP parameters and on cardiovascular outcomes. Unless the intention is to lower nighttime BP, antihypertensive drugs should be taken at a time of day that is convenient, optimizes adherence and minimizes undesirable effects.

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Perspectives

The trials conducted on the efficacy of antihypertensive drugs morning versus evening dosing were small-scale and had heterogeneous outcomes. Particularly, multiple data from a single center have suggested improved 24/48-hour BP control and outcome with evening dosing. A time limited search until May 28, 2022 was conducted using PubMed, EMBASE, and clinicaltrials.gov databases for randomized clinical trials (RCTs) of antihypertensive drug therapies where patients were randomized to morning versus evening dosing. This systematic review and meta-analysis of 72 RCTs evaluates the effect of antihypertensive drugs morning versus evening dosing, which showed that evening dosing of antihypertensive drugs significantly reduced ambulatory BP parameters and lowered cardiovascular events but the effect was mainly driven by trials by Hermida et al. Unless the intention is to specifically lower night-time BP, antihypertensive drugs should be taken at a time of day that is convenient, optimizes adherence, and minimizes undesirable effects.

ARTICLE INFORMATION

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Author Contributions

M.H. Maqsood did the literature search, accessed and verified the data, analyzed and interpreted the data, and wrote the original draft. S. Bangalore conceptualized the study, did the literature search, accessed and verified the data, analyzed and interpreted the data, and wrote the original draft. F.H. Messerli wrote the original draft, conceptualized, and interpreted the data. A.H. Skolnick, J.D. Newman, and J.S. Berger conceptualized and interpreted the data.

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Disclosures

S. Bangalore is on Advisory board for Abbott Vascular, Boston Scientific, Biotronik, Amgen, Pfizer, Merck, REATA, Inari, and Truvic. F.H. Messerli is on the advisory board of Medtronic, Menarini, Ipca, and Krka. The remaining authors have nothing to disclose.

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